

1. Consider the unconstrained optimization problem

$$f(x, y) = 3x^2 + 2y^2 + 4xy - 8x - 6y.$$

Starting from the initial point $x_0 = (0, 0)$,

- (i) (4 mark) perform **two iterations** of the gradient descent method,
- (ii) (2 mark) perform **one iteration** of Newton's method from the same starting point.

2. Consider the constrained optimization problem:

$$\begin{aligned} \min f(x_1, x_2) &= x_1^2 + x_2^2 - 4x_1 - 2x_2 \\ \text{subject to } x_1 + x_2 - 3 &\leq 0, \\ -x_1 &\leq 0. \end{aligned}$$

- (i) (2 mark) State the Karush-Kuhn-Tucker (KKT) conditions for a general constrained optimization problem.
 - (ii) (2 mark) Check whether the point $x^* = (1, 2)$ satisfies the KKT conditions. Comment whether or not x^* is a candidate for optimality.
3. (5 mark) Solve the following transportation problem:

	D_1	D_2	D_3	D_4	Supply
O_1	21	16	25	13	11
O_2	17	18	14	23	13
O_3	32	17	18	41	19
Demand	6	10	12	15	43